

Assignment No 4:

Vijay Mobiles – Festival Sale System

Objective

Vijay Mobiles is conducting a festive sale. Design a Python program using **Lists, Dictionaries, Sets, Comprehensions, and Tuples** to manage inventory, pricing, customer visits, discounts, and billing.

Problem Statement

Create a Python program that performs the following tasks step by step using appropriate data structures.

1. Lists (Creation, Methods, Slicing)

1. Create a list of mobile phones with the following items:
`['iPhone', 'Samsung', 'Vivo', 'Oppo', 'OnePlus']`
 2. A new phone "**Pixel**" arrives in the store.
 - Add it to the existing list using a list method.
 3. The phone "**Vivo**" goes out of stock.
 - Remove it from the list using a suitable method.
 4. The store wants to display the **Top 3 Premium Phones** on the signboard.
 - Use list slicing to display the first three phones.
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2. Dictionaries (Key–Value Pairs, Methods)

1. Create a dictionary to store mobile phone prices.

Example format:

```
{ 'iPhone': 50000, 'Samsung': 40000, ... }
```

2. During the sale, the price of **"iPhone"** is reduced to **₹48,000**.
 - Update the dictionary accordingly.
 3. Add the price for the newly arrived phone **"Pixel"** to the dictionary.
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3. Sets (Unique Elements, Set Operations)

1. You are given two lists of customer names:
 - `morning_visitors`
 - `evening_visitors`
 2. Some customers visited the store both in the morning and evening.
 3. Perform the following operations:
 - Find the **unique customers** who visited the store today (Union).
 - Find the customers who visited **both morning and evening** (Intersection).
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4. Comprehensions (Flash Sale Discount)

1. Apply a **Flash Sale discount of 10%** on all mobile prices.

- Create a new dictionary using **dictionary comprehension** to store discounted prices.
2. Create a list containing **only the discounted prices** using **list comprehension**.
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5. Tuples (Immutable Data Structure)

1. Generate a **tamper-proof bill** for a customer named "**Ravi**".
2. The bill must contain:
 - Customer Name
 - Phone Purchased
 - Final Price
3. Store the bill as a **tuple** and display it.